

Screening of Turner Syndrome with Novel Auxological Criteria Facilitates Early Diagnosis

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Context: Screening criteria for abnormal growth in children are traditionally based on height-for-age, height distance from target height (TH), and change in growth rate. However, there is no consensus on the optimal screening limits.

Objective: Our objective was to first develop new population-based and age-specific cutoffs for these three screening parameters and, second, to validate their performance in screening by using Turner syndrome (TS) as a model.

Design, Patients, and Main Outcome Measure: Reference values for the height distance from TH and growth rate were defined in a total of 14,189 healthy girls aged 0–18 yr with 147,469 height measurements. The best formula for TH calculation was $TH\ SD\ score = 0.79 \times \text{mid-parental height } SD\ score - 0.15$. We also calculated age-specific limits for the distance from TH and limits for growth rate that were freely scalable between age range 0–12 yr. Longitudinal growth data of a cohort including 124 TS girls were evaluated with receiver operating characteristic analysis against the reference population.

Results: The screening accuracy for TS was excellent when we combined all three screening parameters. Sensitivity was 97% and specificity 96% for all TS girls and 100% and 95% for 45,XO TS girls, respectively. The detection rate was 68% for all and 76% for 45,XO TS girls with 99% specificity by the age of 2 yr.

Conclusion: These new population-based screening rules for TS are sensitive and specific, but their use is complex, and therefore their efficient use requires computerization. (*J Clin Endocrinol Metab* 97: E2125–E2132, 2012)

Monitoring of childhood growth has been a part of preventive child healthcare for more than a century (1). Growth monitoring in developed countries is intended to detect childhood illnesses, ideally before any other signs or symptoms of the disease have appeared (2, 3). Important tools in this screening process are certain growth parameters with established cutoff values defining

abnormal growth (4). The two most commonly used parameters are height-for-age [usually expressed as height SD score (SDS) or height percentile] and the distance from target height (TH) [the difference between the child's height SDS (HSDS) and TH SDS, an auxological variable derived from parental heights]. A third growth parameter, the change in growth rate (*i.e.* height velocity), can be

estimated only when longitudinal height measurements are available. Despite the long traditions of growth monitoring, there are few population-based cutoff values defining normal and abnormal growth for the two latter parameters, and there is no consensus about how these parameters should be used in clinical practice (2–5). The first aim of this study was to determine population-based, age-specific cutoff values for the height distance from TH and for the change in growth rate. The second aim was to validate the performance of these cutoff values in clinical practice. We chose Turner syndrome (TS) as a model disease for this study. A short stature is often the only or at least the most obvious clinical feature of TS. Additionally, TS has been regarded as one of the conditions that justify population-based height screening in general (6).

Subjects and Methods

The study was approved by the Ethics Committee of the Pohjois-Savo Health Care District. No contact was made with the study subjects because the data were analyzed anonymously.

Reference population

Mixed cross-sectional and longitudinal growth data of 14,189 healthy girls aged 0–18 yr with 147,469 measurements that were collected for the updated Finnish growth reference comprised the reference population data (7). Children were measured in routine visits at well-baby clinics or school healthcare, and measurements were captured in an electronic patient management system. The quality of growth data gathered at the routine visits has been assessed in Finland, and false measurements, typing errors, missing values, or duplicate recordings are scarce (8). Closer information on the data collection is described elsewhere (7). Subjects of the reference population older than 15 yr and with less than 1.0 cm height gain during the last follow-up year were considered as having reached their adult height (AH) ($n = 3,481$). Mid-parental height (MPH) was the mean of parental heights, corrected for sex (-6.8 cm for girls and $+6.8$ cm for boys) and converted to SDS by using the former Finnish growth reference (from subjects born from 1954–1972), which was considered to better reflect the growth in parents than the new reference (data from subjects born 1983–2009) (7, 9).

Height distance from TH

First, the two commonly used TH formulas were compared to determine which achieved the most precise estimate of the AH in the reference population. The formula developed by Wright and Cheetham (10) is based on the regression line between MPH SDS and AH SDS. The formula developed by Hermanussen and Cole (11), $TH\ SDS = 0.72 \times (\text{paternal height SDS} + \text{maternal height SDS})/2$, has been recommended by the international consensus statement for the diagnosis and treatment of children with idiopathic short stature (12). It is based on correlation coefficients between parental and offspring height SDS and parental height SDS. Second, the differences between HSDS and TH SDS were calculated in the reference population to define age-specific ref-

erence values for HSDS distance from TH SDS using both TH formulas.

Growth rate by age

Calculation of the reference values for normal growth velocity was based on longitudinal growth data of the reference population. Variation of the growth rate over time was based on the fact that the distribution of the change in HSDS in a fixed age interval is normally distributed with zero mean and SD of $\sqrt{[2(1 - r)]}$, in which r is the correlation between the HSDS measurements at the ends of the interval (13). Because the data were not measured at fixed ages, age groups were formed to obtain approximations for the correlations for fixed age intervals. Two different age classifications were used according to the routine measuring schedule adopted in primary care to use the data as exactly as possible: monthly until 1 yr and yearly for whole age period rounding the age to the nearest integer (month or year). Then the correlations were calculated for all the possible pairs of age groups and modeled by regression analysis in which the dependent variable was their Fisher's transformation $z = 0.5 \times \log[(1 + r)/(1 - r)]$. The explanatory variables were the group averages of the difference and average of the pair of ages and/or some transformations of these values, if needed, to better capture the dependence structure between the correlation and age (14). In addition, separate models for two different age classifications were created. These four reported regression models can be used to predict the value of a correlation for any pair of observations using the back transformation formula $r = [\exp(2z) - 1]/[\exp(2z) + 1]$ for the predicted value of z . With the known predicted value of correlation, the predicted value of SD for change can also be calculated by using the above SD formula. The standardized value of change in HSDS is then the actual change divided by its predicted value of SD. In the normal population, it is normally distributed with mean of 0.0 and SD of 1.0. To depict the normal variation of the growth rate over time, estimates of SD curves for different time intervals were plotted from each model. Goodness of fit for the regression model was assessed by creating different residual plots and plotting fitted curves for SD together with corresponding data points. Parameter estimates were used to define formulas for growth rate reference values that are presented in Supplemental Table 1 and Supplemental Appendix 1 (published on The Endocrine Society's Journals Online web site at <http://jcem.endojournals.org>) for girls aged from 0–1 yr and from 0–12 yr. According to these formulas, HSDS change over time was standardized by age distance and mean age between two measurements, and cutoffs for changes in growth were defined (see Supplemental Table 2). These population-based cutoff values for growth rate are scalable for age without limitation of fixed time intervals.

TS cohort

Longitudinal height data including 2184 measurements for 136 girls with TS (born between 1978 and 2009) were collected from three Finnish University Hospitals (Helsinki, Kuopio, and Tampere). TS girls were identified from the hospitals' patient registry by International Classification of Diseases (ICD-10) codes (Q96.0–9). Clinical data including gestational age at birth, birth size, longitudinal growth data, parental heights, karyotype, the date of TS diagnosis, associated disorders, treatments, and medications were recorded. Twelve TS girls were excluded from the study population: three girls with type 1 diabetes, one with

a malignant tumor, one with hyperinsulinism, and seven because of missing parental heights. The remaining TS cohort comprised 124 (91.2% of the original cohort) girls and their 2020 height measurements (Table 1). All the TS girls were measured in the standard way with calibrated equipment by specially trained nurses in scheduled, routine visits at well-baby clinics, school healthcare, or control visits at the hospital outpatient clinics (8). Potentially false measurements, typing errors, missing values, or duplicate recordings were detected in the longitudinal height data using scatter plots and then either corrected or excluded. Height data only until the start of any growth-promoting therapy were used.

Statistical analyses

Paired-samples *t* test was used to test the difference between the two TH formulas. Validation of the three growth screening parameters [height-for-age in comparison with the population mean (HSDS rule); TH, *i.e.* the maximal HSDS distance from TH SDS (TH SDS rule); change in growth rate, *i.e.* the maximum change in HSDS over time (Δ HSDS rule)] were evaluated one by one and in combination. First, the individual probability for abnormal growth screening (using HSDS, TH SDS, Δ HSDS, or the combination rule) were calculated for each girl in the TS cohort and the reference population using logistic regression.

Second, these probabilities were used to assess the diagnostic accuracy of growth screening in the whole TS population and in the subpopulation with the 45,XO karyotype. The diagnostic accuracy was depicted by area under the curve (AUC), values from receiver operating characteristic (ROC) curves against the reference population. Third, the optimal cutoff points (OCP) for each screening rule were defined so that sensitivity and specificity of the rule obtained the maximal values (OCP was chosen at the point of maximum of sensitivity² + specificity²). Cumulative diagnostic performance of each screening rule was also evaluated by age.

Data were analyzed using R statistical software and SPSS software (version 19; IBM Corp., Armonk, NY).

Results

Population-based cutoffs for height-for-age, height distance from TH, and growth rate

The modified Wright and Cheetham (10) formula in the reference subpopulation with known AH (*n* = 3481) resulted in the following TH formula: TH SDS = 0.79 ×

TABLE 1. Characteristics of the 124 girls with TS

	45,XO karyotype	Other karyotypes	All
Sample size (<i>n</i>)	60	64	124
Number of measurements (count)	980	1040	2020
Number of measurements per girl	16.3 (7.14)	16.3 (6.95)	16.3 (7.01)
HSDS, ^a all measurements	−2.24 (0.80)	−2.07 (1.08)	−2.15 (0.96)
Weight for height SDS, ^a all measurements	−0.02 (1.05)	0.01 (1.40)	−0.01 (1.25)
BMI for age SDS, ^a all measurements after 2 yr	0.02 (0.88)	0.05 (1.32)	0.03 (1.14)
Height SDS ^b (TS reference), all measurements	0.45 (0.74)	0.70 (1.04)	0.58 (0.91)
Gestational age (wk) (<i>n</i> = 91) ^c	38.9 (1.71)	38.9 (2.34)	38.9 (2.04)
Birth height (cm) (<i>n</i> = 91)	47.3 (1.94)	47.6 (2.99)	47.5 (2.53)
Birth weight (kg) (<i>n</i> = 91)	2.99 (0.49)	3.03 (0.65)	3.01 (0.57)
Birth height SDS ^d (<i>n</i> = 91)	−1.23 (0.98)	−0.90 (1.32)	−1.07 (1.17)
Birth weight SDS ^d (<i>n</i> = 91)	−1.04 (1.17)	−0.96 (1.25)	−1.00 (1.20)
Paternal height (cm)	180.0 (6.31)	177.7 (7.23)	178.8 (6.87)
Paternal height SDS ^e	0.17 (0.95)	−0.17 (1.08)	−0.01 (1.03)
Maternal height (cm)	165.7 (5.3)	165.4 (6.00)	165.6 (5.64)
Maternal height SDS ^e	0.07 (0.92)	0.02 (1.04)	0.05 (0.98)
MPH (cm)	172.9 (4.42)	171.6 (4.76)	172.2 (4.62)
MPH SDS ^e	0.12 (0.71)	−0.08 (0.76)	0.02 (0.74)
Age at diagnosis (yr) (<i>n</i> = 118) ^f [median (range)]	0.0 (0–17.0)	5.3 (0–15.9)	3.6 (0–17.0)
Perinatal diagnose (count)	31	12	43
Karyotype [<i>n</i> (%)]			
45,XO	60 (100)		60 (48)
45,XO and 46,XX		13 (20)	13 (11)
45,XO and [46,X,iX or 46,X,idi(X)]		10 (16)	10 (8)
46,X, iX or 46,X, idi(X)		5 (8)	5 (4)
45,XO/47,XXX		4 (6)	4 (3)
Other		32 (50)	32 (26)

Unless indicated otherwise, results are shown as mean (SD).

^a Finnish growth reference 1983–2008 (7), all measurements.

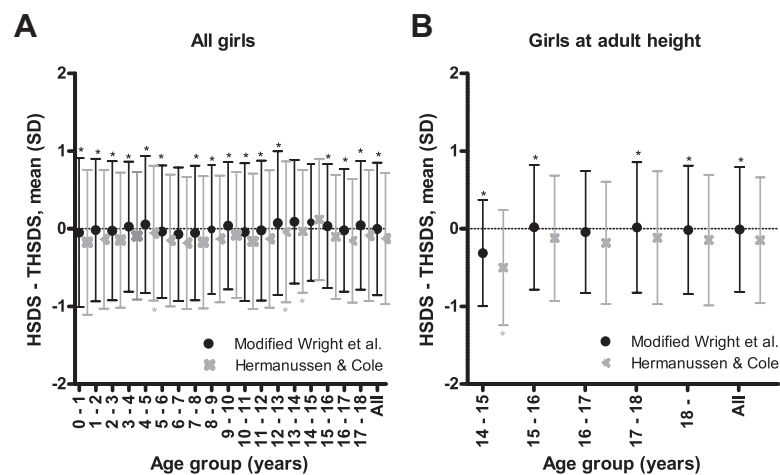
^b Northern European growth reference for TS (22), all measurements.

^c Not found for 33 TS girls.

^d New Finnish growth references for birth size (unpublished study).

^e Finnish growth reference 1959–1971 (9).

^f Not found for six TS girls.



*Height SDS difference from TH SDS is zero ($P < 0.05$)

FIG. 1. Comparison of the modified TH formula by Wright and Cheetham (10) and the formula of Hermanussen and Cole (11) in the reference population. Observed HSDS difference from the target TH SDS is shown for all girls in the reference population (A) and for the girls that had reached their AH (B) by age groups. *, Lack of difference from zero ($P < 0.05$). See also Supplemental Appendix 2 and Supplemental Table 2 for the mathematical algorithms and age-specific cutoff limits for the TH SDS rule.

MPH SDS -0.15 (Supplemental Fig. 1 and Supplemental Appendix 2).

This was compared with the Hermanussen and Cole formula (11) with respect to AH SDS and HSDS at ages

for further use.

Age-specific cutoff values for HSDS distance from TH SDS could be calculated from the reference values shown in Fig. 1A. These cutoffs at the specificity of 99% ranged from 1.9–2.2 SDS in age groups of 2–12 yr (see Supplemental Appendix 2 and Supplemental Table 2).

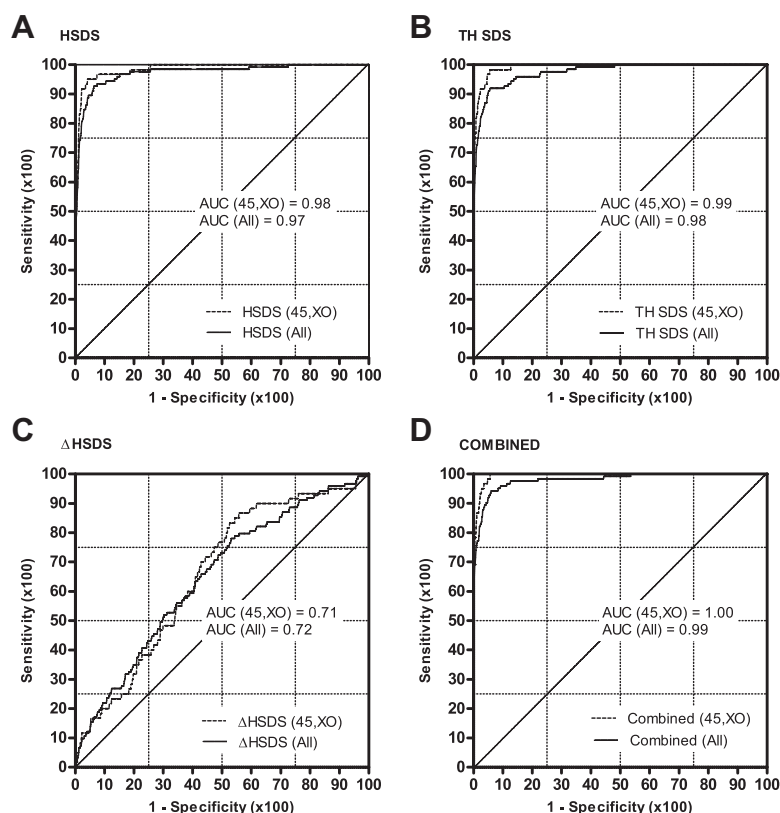


FIG. 2. ROC curves for the three growth screening rules (A–C) and their combination (D) in the TS population ($n = 124$, solid lines) and in a subsample of TS girls with karyotype 45,XO ($n = 60$, dashed lines): A, absolute HSDS rule; B, TH SDS rule; C, Δ HSDS rule; D, combination of these three rules (HSDS, TH SDS, or Δ HSDS) (D). AUC is shown separately for every rule.

0–18 yr. The mean (SD) difference between HSDS and TH SDS was 0.00 (0.85) in all girls of the reference population and -0.01 (0.80) in girls who had achieved their AH when using the modified formula by Wright and Cheetham (10) and -0.13 (0.84) and -0.15 (0.81), respectively, when using the formula of Hermanussen and Cole (11) ($P \leq 0.05$) (Fig. 1). In addition, the mean difference between actual HSDS and TH SDS was significantly smaller when the modified Wright and Cheetham (10) formula was used, and this was true in virtually all age groups from 0–12 yr ($P \leq 0.05$), especially in short children. Because the formula by Wright and Cheetham (10) appeared to be more concordant with the AH SDS and childhood HSDS than that of Hermanussen and Cole (11), it was selected

Screening accuracy for TS

With either HSDS rule or TH SDS rule, screening accuracy was slightly better for 45,XO TS girls than in the whole TS cohort. For HSDS rule, the AUC value was 0.98 (95% CI = 0.97–1.00) in 45,XO TS girls and 0.97 (0.96–0.99) in all TS girls, and for TH SDS rule, 0.99 (0.99–1.00) and 0.98 (0.96–0.99), respectively (Fig. 2). In contrast, screening accuracy for Δ HSDS rule was slightly better for all TS girls than for 45,XO girls with AUC of 0.72 (0.68–0.76) vs. 0.71 (0.66–0.77). Δ HSDS rule displayed a poorer screening accuracy than HSDS or TH SDS rule alone. Combining the three rules resulted in an excellent screening accuracy with AUC of 1.00 (0.99–1.00) for 45,XO TS girls and 0.99 (0.98–1.00) for all TS girls, respectively.

The OCP for HSDS rule was observed at -1.92 SDS with a sensitivity of 93% and specificity of 93% in the whole TS group, and -2.13 SDS with a sensitivity of 95% and specificity of 96% in the 45,XO group. The OCP

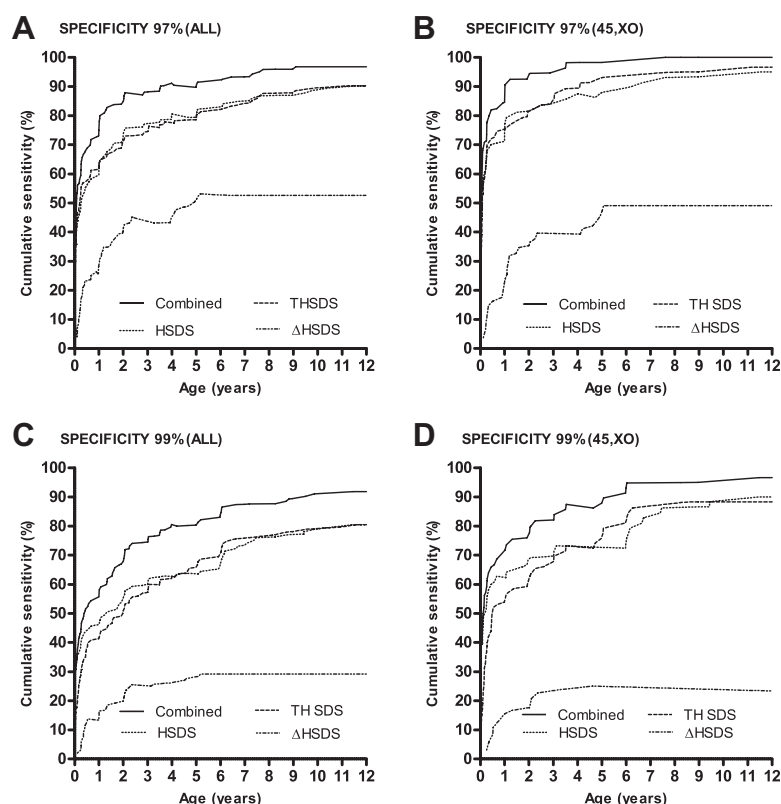


FIG. 3. Cumulative sensitivities of four growth screening rules from 0–12 yr: HSDS rule, TH SDS, Δ HSDS, and combination of these three in the whole TS group (A and B) ($n = 124$) and in the subsample of TS girls with karyotype 45,XO (C and D) ($n = 60$) at specificity levels of 97 and 99%, respectively.

for TH SDS rule and Δ HSDS rule were -2.03 (sensitivity and specificity of 92 and 94%) and -0.40 (96 and 32%) for all and -2.65 (98 and 94%) and -0.40 (97 and 32%) for 45,XO TS girls, respectively. When the combination rule was used, OCP (HSDS rule -2.43 , TH SDS rule -1.60 , and Δ HSDS rule -1.44) was observed at sensitivity and specificity of 97 and 96% for all and 100 and 95% for 45,XO TS girls.

At all ages, the combination rule provided the highest sensitivity for all TS girls and also for the subgroup of 45,XO girls (Fig. 3). By using the combination rule, 85% of all and 93% of 45,XO TS girls were detected by the age of 2 yr with 97% specificity and 68% of all and 76% of 45,XO TS girls with 99% specificity, respectively. Additionally, all the 45,XO girls were detected by the age of 7 yr with 97% specificity and 97% of 45,XO girls by the age of 12 yr with 99% specificity.

To some extent, all three screening rules complemented each other, because some of the girls were detected by a single rule only. The first alert was triggered by the HSDS rule in 20%, by the TH SDS rule in 24%, and by the Δ HSDS rule in 14% of the TS girls, respectively. Two or three rules were triggered in 42% of TS girls at the same time. In other words, removing any of the three rules from the combination resulted in lower

sensitivity than could be obtained by the combination of the three rules. However, the Δ HSDS rule did not improve the screening performance as much as the HSDS and TH SDS rules alone.

Cumulative sensitivities of the HSDS and TH SDS rules and the combination of the three rules increased until 12 yr of age, whereas with the Δ HSDS rule, the highest sensitivity was obtained already by the age of 5 yr.

Examples of growth monitoring using the three screening rules (HSDS, TH SDS, and Δ HSDS) in eight TS girls are presented in Fig. 4. The first alert generated by each of the three rules is marked in the figure. Overall, growth screening alerts were observed in 112 TS girls (91%) from 0–12 yr.

Discussion

Growth monitoring is a fundamental part of primary healthcare in children, but the performance of various growth screening rules has remained inadequately explored. A recent systematic review on growth monitoring for short stature showed that OCP for abnormal growth are practically lacking (2). One major strength of our study was establishment of population-based growth screening rules, which were developed using a large, contemporary height data set of healthy girls.

In this study, we performed an extensive methodological exercise to optimize population-based growth screening. First, age-specific normal values were calculated for the height distance from TH. Our data showed that height distance from TH in population is age dependent and should be taken into account in growth monitoring. Second, we developed a method based on change in HSDS to assess changes in growth rate over time between age points from 0–12 yr. As far as we are aware, this is the first study to define population-based, age-specific reference values for HSDS distance from the TH SDS and limits for the changes in growth rate that are scalable freely between any two age points. Ultimately, we validated our new population-based growth screening rules for HSDS, height distance from TH SDS, and HSDS changes in a retrospective clinical data set of 124 girls with TS. Using these three growth parameters, abnormal growth in TS was detected with excellent accuracy already at a young age.

In this study, TS was used as a model disorder because of the fact that short stature is often the only or the most obvious clinical sign. However, growth monitoring pro-

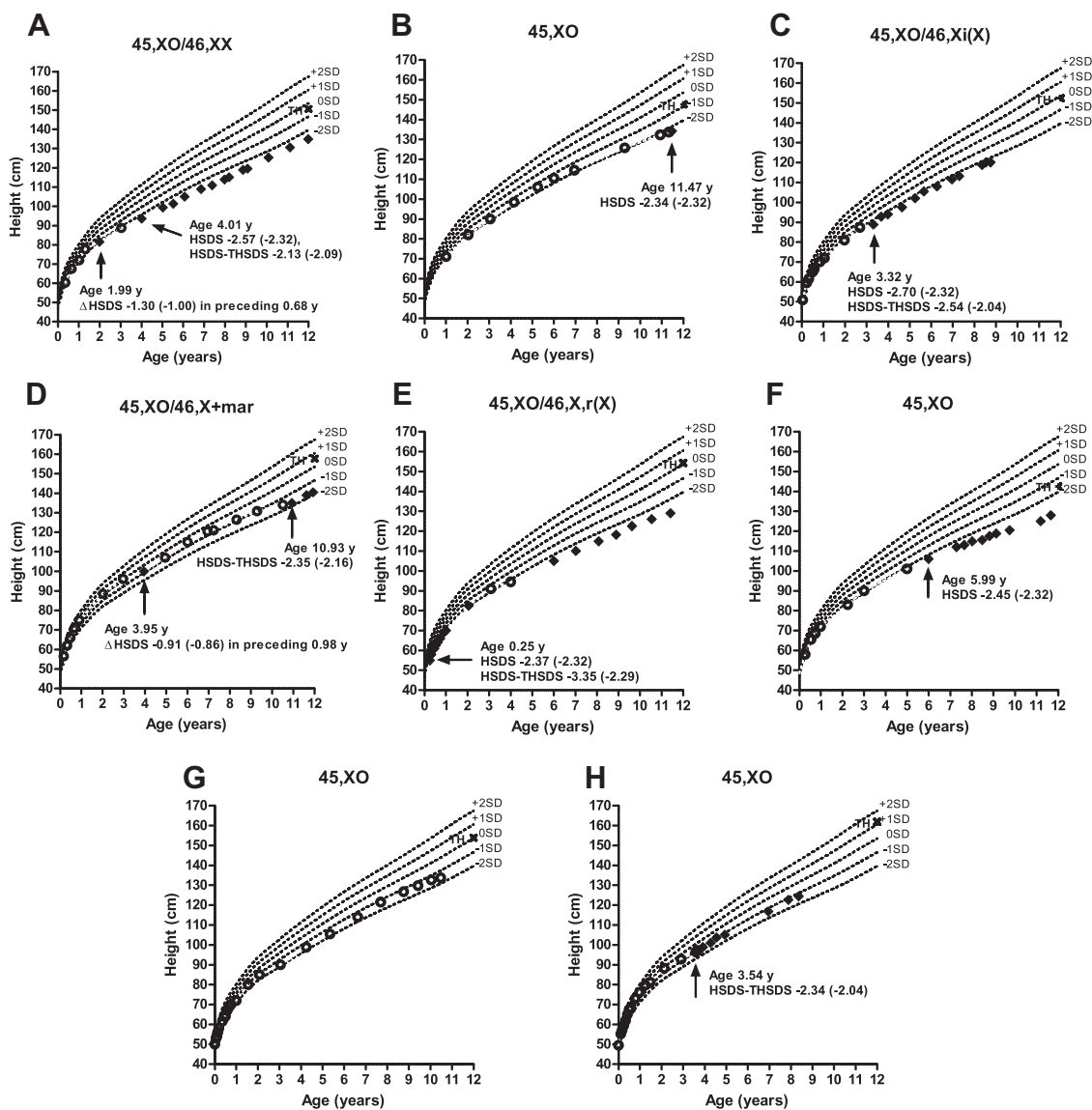


FIG. 4. Height-for-age curves of eight girls (A–H) with TS (karyotypes shown above each graph). Data were screened at specificity of 99% for all three screening rules: HSDS, HSDS distance from TH SDS, or change in HSDS over time (Δ HSDS). The first alert triggered by each of the rules is marked with an arrow; the rule generating the alert is specified together with cutoff value of the rule in brackets. Open circles denote normal measurements with no alerts and diamonds measurements with an alert triggered by one or more of the three screening rules.

grams aim at identifying not only TS but also a variety of other disorders affecting growth. Grote *et al.* (3) have shown that the growth patterns of the most prevalent growth disorders (TS, GH deficiency, celiac disease, and cystic fibrosis) vary by age and the screening rule to be used, but overall TS appears to be an optimal target group for screening. However, additional studies in other disorders than TS are still warranted.

An ideal growth monitoring program should have a high sensitivity to detect as many abnormally growing children as possible, with a specificity that does not produce too many unnecessary referrals. However, there is always a trade-off between sensitivity and specificity. For the HSDS cutoff (or height percentile cutoff) values, the

specificity is predefined by the growth references without any knowledge of the sensitivity for each of the detectable growth disorders. In the previous study of van Buuren *et al.* (4), 70% of TS girls were detected by the HSDS rule with a specificity of 93%. In contrast, we were able to detect as many as 93% of TS girls with the same specificity by using the HSDS rule. The difference can be explained by two factors. The TS girls in our cohort were born between 1978 and 2009, as were the majority of the children in the reference population (7), whereas in the Dutch study, TS girls were older than children in the reference data (born between 1968 and 1996 in comparison with reference girls born between 1989 and 1990). Furthermore, the reference children were on average 0.31 SDS shorter than the Dutch reference population (4).

Height distance from TH, in comparison with HSDS, is believed to improve sensitivity without significantly reducing specificity, because the SD for TH is narrower than the SD for height (15). Several methods have been proposed for calculating TH (10, 11, 15, 16). In our study, the TH SDS formula of Wright and Cheetham (10) was very accurate in predicting the AH and was also consistent with HSDS in all age ranges in the population sample.

In the study of van Buuren *et al.* (4), height distance from TH in the screening for TS was assessed as well, although in their study, a large proportion of maternal or paternal heights were imputed due to missing data. Screening of TS with the TH rule in our study produced similar performance as that reported by van Buuren *et al.* (4). In another Dutch study reported by Grote *et al.* (3), it was found that the use of TH was superior to HSDS only in growth screening. Therefore, parental heights should be systematically recorded for each child, and TH-based screening may then become useful, especially in multiethnic populations when population-specific growth references are not available. Gozzi *et al.* (17) reported that short parents overestimate their height, and therefore actual measurement of parental heights if possible is preferable.

Change in growth rate is almost an unexplored method in the screening of growth disorders, and population-based reference values for normal changes are virtually nonexistent (18, 19). One advantage in the method reported in this study is that the cutoff values for abnormal growth rate can be defined for any age range between 0 and 12 yr. Nevertheless, use of growth rate (Δ HSDS rule) is highly dependent on the accuracy of height measurements.

In our experience, almost 40% of growth disorders can be identified based on an abnormal growth rate before the child is abnormally short or tall (unpublished observation). In one previous study, the combination of height-for-age, height distance from TH, and change in growth rate has been reported to detect children with short stature with good sensitivity and acceptably high specificity (3). Our result in the screening of TS substantiates these findings, although TS is a growth disorder with a prenatal origin, and thus the main growth feature is a short stature for age or abnormal distance of height from TH. Therefore, screening by using the changes in growth rate is probably more accurate in disorders that typically manifest as decrease in HSDS or height percentile, *e.g.* GH deficiency, hypothyroidism, and celiac disease.

Growth in TS girls typically is affected already *in utero* and in infancy, but nonetheless, in two previous studies, only about one third of TS girls had been identified by mid-childhood (20, 21). Our results on cumulative sensitivity by age revealed that TS girls can be potentially de-

tected much earlier, as many as 80% (specificity 99%) by the age of 5 yr by systematic population-based growth monitoring and by following the screening rules. Furthermore, karyotype should be examined for girls with typical clinical findings for TS (*e.g.* lymphedema at birth, webbed neck, or cubitus valgus), especially if they are short.

In conclusion, an optimal growth monitoring program requires standardized equipment, well-trained nurses, an adequate population-based reference, and evidence-based screening rules and referral criteria approved by pediatric growth specialists. The establishment of population-based, validated cutoff values for height-for-age, height distance from TH, and changes in growth rate would represent a crucial step toward evidence-based growth monitoring and should be implemented in the growth monitoring programs. The screening algorithms such as the cutoff values for growth rate are complex, and therefore, their efficient use requires computerization. The integration of screening algorithms into contemporary electronic patient management systems could be an interesting way to screen for growth disorders in the future.

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